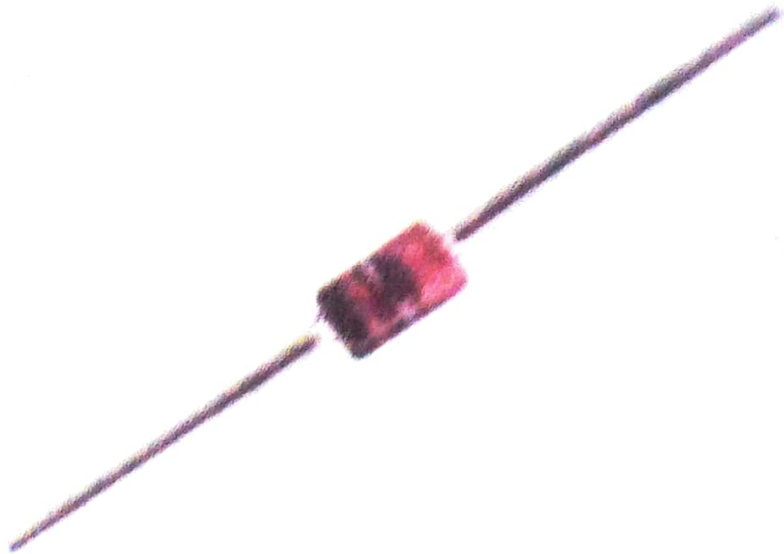
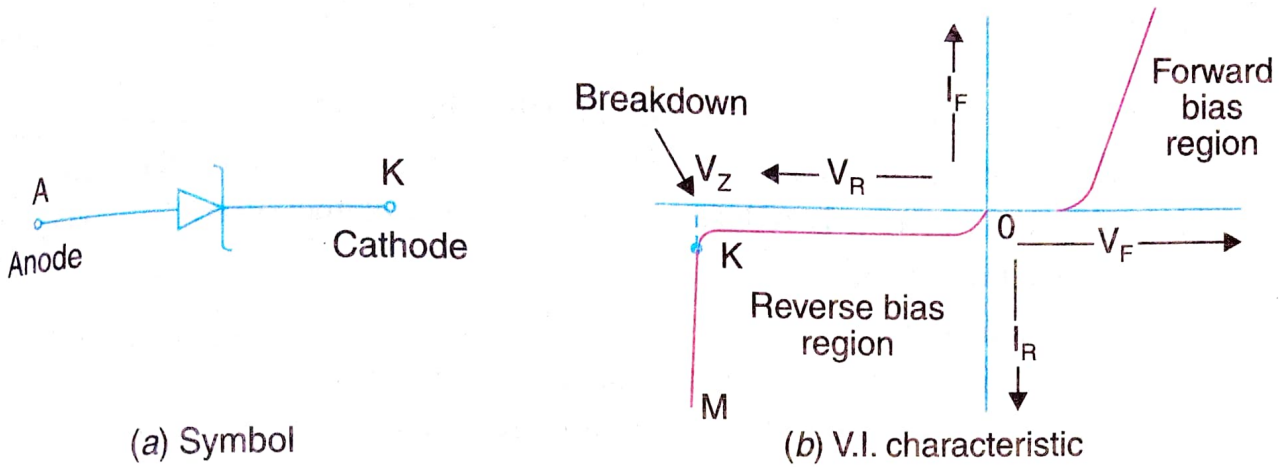
13.2. Zener Diode

A zener diode is also called a voltage-reference, voltage regulator or breakdown diode. Like a rectifier diode, it is also important in many power applications. Fig. 13.1 (*a*) shows a schematic symbol for a zener diode. It will be interesting to know that the line at



Zener diode

the end of the arrow looks like the letter Z.



(*a*) Symbol (*b*) V.I. characteristic

Fig. 13.1. Zener diode.

The zener diode is a silicon PN junction device, which differs from a rectifier diode, in the sense, that it is operated in the reverse break down region. The breakdown voltage of a zener diode is set by carefully controlling the *doping level* during manufacture. As discussed in the last chapter, when a reverse voltage across a diode is increased, a critical voltage called breakdown voltage is reached at which the reverse current increases sharply as shown by the curve *KM* in Fig. 13.1 (*b*).

The reverse breakdown of a PN junction may occur either due to *avalanche* or *zener effect.* As discussed in chapter on PN-junctions the avalanche breakdown occurs, when the accelerated free electrons acquire sufficient energy to ionize a lattice atom by bombardment. The additional free electrons, created in this way, are accelerated by the reverse field causing more and more ionization. The multiplication of the number of free carriers causes the reverse current to increase rapidly.

The zener breakdown occurs when the electric field across the junction, produced due to reverse voltage, is sufficiently high. This electric field exerts a force on the electrons in the outer most shell. This force is so high that the electrons are pulled away from their parent nuclei and become free carriers. This ionization, which occurs due to the electrostatic force of attraction, is known as zener effect. It causes an increase in the number of free carriers and hence an increase in the reverse current.

The zener diodes, with breakdown voltages of less than 6 V, operate predominantly in zener breakdown. Those with breakdown, voltages greater than 6 V, operate predominantly in avalanche breakdown. Strictly speaking, the first one should be called a zener diode and the second an avalanche diode. But in actual practice, both the types are called zener diodes.

diode. But

13.3. Reverse Characteristic of a Zener Diode

We have just discussed that a zener diode is operated only in the reverse-bias region. That is why we shall examine its characteristic in this region only. Fig. 13.2 shows the reverse portion of the V-I characteristic of the zener diode. It may be noted from this figure that as the reverse voltage (generally written as V_R) is increased, the reverse current (usually called zener current, I_Z) remains negligibly small up to the 'knee' of the curve (point K in Fig. 13.2). At this point, the effect of breakdown process begins. From the bottom of the knee, the breakdown voltage (also called zener breakdown voltage or simply zener voltage V_Z) remains essentially constant. This ability of a diode is called regulating ability and is an important feature

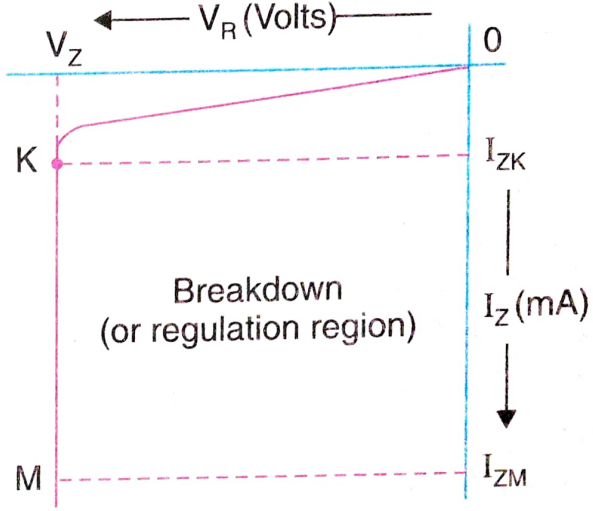


Fig. 13.2. Reverse characteristic of a zener diode.

of a zener diode. It maintains, an essentially a constant voltage across its terminals over a specified range of zener current values. Following two points are important from the characteristics of a zener diode:

1. There is a minimum value of zener current called breakover current designated as I_{ZK} or $I_{Z\ (min)}$ which must be maintained in order to keep the diode in breakdown (or regulation) region. When the current is reduced below the knee of the curve, the voltage changes drastically and the regulation is lost.
2. There is a maximum value of zener current designated as I_{ZM} or $I_{Z\ (max)}$ above which the diode may be damaged. The value of this current is given by the maximum power dissipation of the zener diode. As long as the maximum power dissipation is not exceeded, the diode will not be damaged. It will come out of the breakdown region, when the applied reverse voltage is reduced below the breakdown voltage.

13.4. Zener Diode Specifications

The zener diodes are generally specified in terms of four factors namely zener voltage (V_Z) maximum power dissipation (P_{ZM}, $P_{D\ (max)}$ or $P_{Z\ (max)}$), breakover current (I_{ZK}) and zener resistance (r_Z). They are available with breakdown voltages ranging from 1.8 V to 2000 V.

The power dissipation of a zener diode is the product of breakdown voltage (V_Z) and reverse current (I_Z). Mathematically, the power dissipation,

$$P_Z = V_Z \times I_Z$$

The maximum value of power dissipation, which a zener can dissipate, without failure, is called power rating and is designated by P_{ZM}. The zener diodes are available with power rating from 150 mW to 50 W. Generally, the data sheets of zener diodes include the maximum current, a zener can handle without exceeding its power rating. The value of this maximum current is given by the relation,

$$I_{ZM} = \frac{P_{ZM}}{V_Z}$$

where

P_{ZM} = Power rating of a zener diode, and

V_Z = Breakdown voltage.

Fig. 13.3 (a) and (b)

13.5. Zener Diode Equivalent Circuit

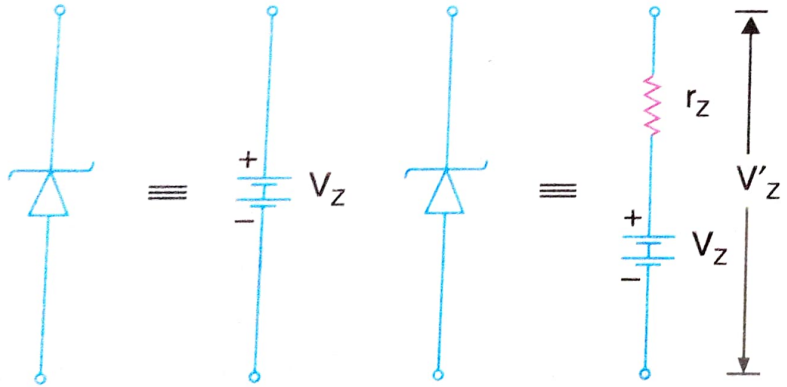
Fig. 13.4 (*a*) shows an ideal approximation of a zener diode in reverse breakdown. It shows that a zener diode is simply equivalent to a battery having a voltage equal to the zener voltage (V_Z) Fig. 13.4 (*b*) shows a practical equivalent circuit of a zener diode. This circuit shows that a zener diode is equivalent to a battery with voltage, (V_Z) in series with a resistance (r_Z) called zener resistance. The zener resistance is also known as *dynamic resistance* or *a.c. resistance.*

The presence of resistance (r_Z) indicates that the reserve V-I characteristic of a zener diode is not ideally vertical but it is slightly tilted as shown in Fig. 13.5. The value of zener resistance may be obtained from the V-I characteristic as discussed below:

Consider any two point P and Q on the reverse V-I curve as shown in Fig. 13.5.

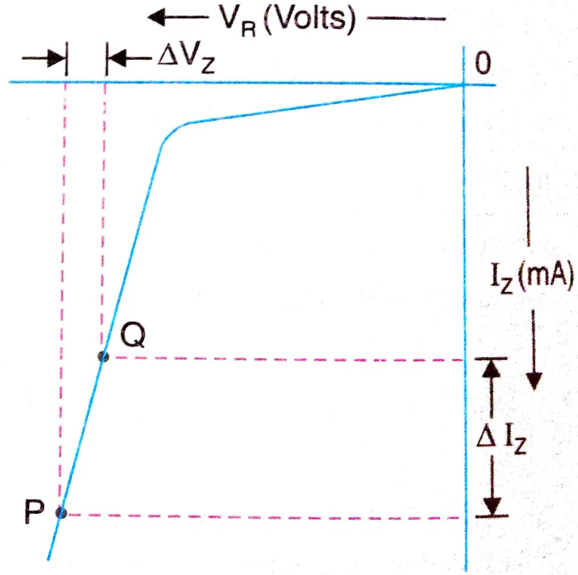
Let ΔV_Z = Change in the values of zener voltage between the points P and Q, and

ΔI_Z = Change in the values of zener current at the corresponding points.



(*a*) Ideal zener equivalent diode circuit.

(*b*) Practical zener diode equivalent circuit.



Reverse V.I characteristic curve illustrating zener resistance.

We know that zener resistance is given by the relation,

$$r_Z = \frac{\Delta V_Z}{\Delta I_Z}$$

For a perfectly vertical breakdown curve, the value of zener resistance is zero. But in actual practice, its value may vary from few ohms to several hundred ohms. The variation in its value depends upon the particular zener voltage and the value of operating current. It may be noted that if the zener current flows through the zener diode in the direction as shown in Fig. 13.5, then voltage drop (V_Z) across the zener resistance has the same polarity as that of a battery. In this case, the voltage across the terminals *A* and *B*,

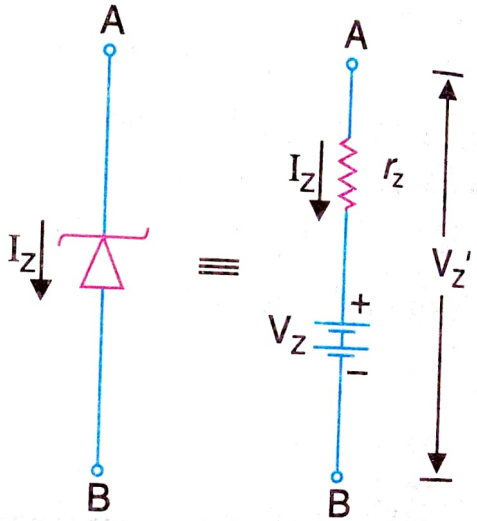


Fig. 13.6. Terminal voltage of a zener diode.

$$V_Z' = I_Z \cdot r_Z + V_Z$$

If the value of r_Z is very small, then the voltage across the terminals *A* and *B*,

$$V_Z' = V_Z$$

Example 13.3.

When the reverse current in a particular zener diode increases from 20 mA to 30 mA, the zener voltage changes from 5.6 V to 5.65 V. What is the resistance of device?

Solution.

Given: $\Delta I_Z = 30 - 20 = 10$ mA $= 10 \times 10^{-3}$ A and $\Delta V_Z = 5.65 - 5.6 = 0.05$ V.

We know that resistance of the zener diode,

$$r_Z = \frac{\Delta V_Z}{\Delta I_Z} = \frac{0.05}{10 \times 10^{-3}} = 5\ \Omega$$ **Ans.**

13.6. Zener Diode Applications

Though the zener diodes have a number of applications, yet the following are important from the subject point of view :

1. As a voltage regulator.
2. As a fixed reference voltage in transistor biasing circuits.
3. As peak clippers or limiters in waveshaping circuits.
4. For meter protection against damage from accidental applications.

20.5. Zener Diode Shunt Regulator

Fig. 20.4 shows a circuit of zener diode shunt regulator. Since the zener is connected in parallel (or shunt) with the load, therefore the circuit is known as shunt regulator. A resistance, (R_S) is connected in series with the zener to limit current in the circuit. Therefore the resistance R_S is also known as series current limiting resistor.

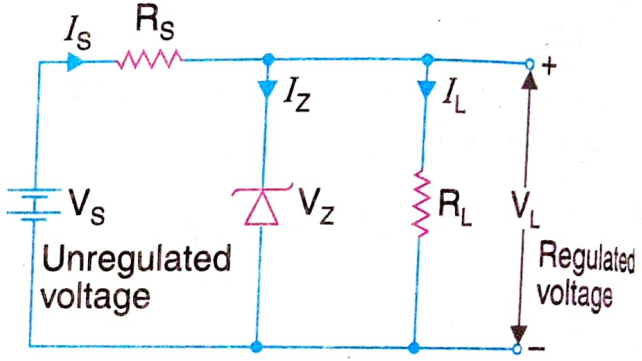


Fig. 20.4. Zener diode shunt regulator.

The output voltage (V_L) is taken across the load resistance (R_L). For proper operation, the input voltage (V_S) must be greater than the zener voltage (V_Z). This ensures that zener operates in the reverse breakdown region.

The input current (*i.e.*, current through the limiting resistor) is given by the relation,

$$I_S = \frac{V_S - V_Z}{R_S}$$

where V_S = The d.c. input voltage to the regulator circuit, and

V_Z = The zener voltage.

The ideal zener diode acts as a constant voltage source of voltage (V_Z). However, a practical zener diode has a finite value of resistance called zener resistance (r_Z). Because of the zener resistance, there is a voltage drop across it which is equal to $I_Z \cdot r_Z$. Therefore the voltage across the terminals of the zener diode (which is equal to load voltage),

$$V_L = V_Z + I_Z \cdot r_Z \quad ...(i)$$

If the zener resistance is negligible, then the load voltage,

$$V_L = V_Z \quad ...(ii)$$

and the current through the load resistance, is given by the relation,

$$I_L = \frac{V_L}{R_L} \quad ...(iii)$$

We know that input current is the sum of zener current and load current *i.e.*,

$$I_S = I_Z + I_L \quad \text{or} \quad I_Z = I_S - I_L$$

Example 20.4

Fig. 20.5 shows the circuit of a zener diode shunt regulator.

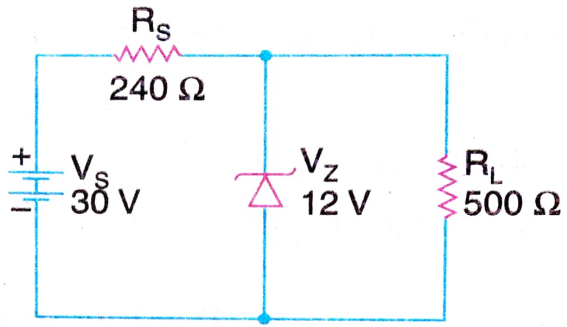


Fig. 20.5.

Find (a) the load voltage (b) voltage drop across series resistance and (c) current through the zener diode.

Solution.

Given : $V_S = 30$ volts; $R_S = 240\ \Omega$; $V_Z = 12$ volts and $R_L = 500\ \Omega$.

(a) *Load Voltage*

We know that the load voltage

$$V_L = V_Z = 12 \text{ V } \textbf{Ans.}$$

(b) *Voltage drop across series resistance*

We know that current through series resistance R_S.

$$I_S = \frac{V_S - V_Z}{R_S} = \frac{30 - 12}{240} = 0.075 \text{ A}$$

and voltage drop across series resistance R_S.

$$= I_S \cdot R_S = 0.075 \times 240 = 18 \text{ V } \textbf{Ans.}$$

(c) *Current through zener diode*

We also know that current through R_L,

$$= I_L = \frac{V_L}{R_L} = \frac{12}{500} = 0.024 \text{ A}$$

and current through zener diode,

$$I_Z = I_S - I_L = 0.075 - 0.024 = 0.051 \text{ A } \textbf{Ans.}$$